VAWT Aerodynamic Data — Project README

Data Sources and Folder Organisation

Overview

This folder contains aerodynamic coefficient data for NACA symmetric airfoil profiles used in Vertical Axis Wind Turbine (VAWT) analysis. It is split into two main subfolders: **Experimental Data** and **Synthesized Data**.

Folder Structure

1. Experimental Data

This folder contains data digitised from the wind-tunnel experiment plots in the Sheldahl and Klimas (1981) paper. The paper showed the measurements as graphs but never listed the numbers in a table. The data points were therefore extracted from the graphs using WebPlotDigitizer.

It covers four airfoil profiles: **NACA 0009, 0012, 0012H, and 0015**. Each profile has its own subfolder with the following contents:

- **Lift Coef vs Alpha** — Lift coefficient (C_L) against angle of attack (α) from 0° to 180°, at each tested Reynolds number. Contains `.csv` data files, a MATLAB plotting script, and a `figures/` subfolder with the output plots.
- **Drag Coef vs Alpha** — Same structure as above, for drag coefficient (C_D) across the full angle-of-attack range.
- **Axial Force Coefficient vs Alpha** — Axial force coefficient (C_A) against α at each Reynolds number. Available for **NACA 0012, 0012H, and 0015** only (not 0009). Contains `.csv` files, a MATLAB script, and output figures.
- **Moment Coef vs Alpha** — Pitching moment coefficient (C_M) about the quarter-chord point, against α. Contains `.csv` files, a MATLAB script, and output figures.
- **Lift and Drag vs Alpha** — C_L and C_D plotted together on the same axes for direct comparison. Contains a MATLAB script and output figures.
- **Lift vs Drag Polar** — Lift-drag polar plots (C_L vs. C_D). Contains a MATLAB script and output figures.

In addition, there are three shared subfolders that sit outside the individual airfoil folders and cover all profiles together:

- **Lift Coefficients — Increasing & Decreasing Alpha** — C_L data for both increasing and decreasing angles of attack, showing any hysteresis in the measurements. Covers NACA 0009 (Fig. 3), 0012 (Fig. 4), 0012 at high Reynolds numbers (Fig. 5), 0012H (Fig. 6), and 0015 (Fig. 7). Each profile has its own subfolder with `.csv` files and a MATLAB script.
- **Drag Coefficient — Small Angles of Attack** — C_D data in the low-angle range ($-10°$ to 10°). Covers NACA 0009 (Fig. 13), 0012 (Fig. 14), 0012 at high Reynolds numbers (Fig. 15), 0012H (Fig. 16), and 0015 (Fig. 17). Each profile has its own subfolder with `.csv` files and a MATLAB script.
- **Full Range Datasets** — Three Excel workbooks with the complete collated datasets across all four profiles:

- `Full Range Lift Coefficients.xlsx`
- `Full Range Drag Coefficients.xlsx`
- `Full Range Moment Coefficients.xlsx`

2. Synthesized Data

This folder contains data from the tabulated dataset in the 1981 paper. This dataset covers a much wider range of Reynolds numbers than the experiments. Sheldahl and Klimas produced it by combining the wind-tunnel results with numerical models (the Viterna–Corrigan post-stall model and the Eppler code) to fill in Reynolds numbers that were not tested directly.

It covers five profiles: **NACA 0012, 0015, 0018, 0021, and 0025**. The three thicker profiles (0018, 0021, 0025) were not tested in the wind tunnel — their data was generated entirely through the synthesiser code.

Each profile has its own subfolder containing interactive HTML plot files, one per Reynolds number (e.g. `Re_360k.html`, `Re_1M.html`). An `index.html` file in each subfolder links to all plots for that profile. A parent `index.html` at the top of this folder links to all profiles.

Also included at the top level of this folder:

- `Sheldahl_Kilmas_1981_Excel_Sheets_Plots.m` — The MATLAB script used to generate the HTML plots from the tabulated data.
- `Sheldahl_Klimas_1981_converted.xlsx` — The tabulated data from the 1981 paper, converted into an Excel workbook.

Data Sources

1. Sheldahl and Klimas (1981) — Primary Source

The main dataset for this project comes from:

> R. E. Sheldahl and P. C. Klimas, *"Aerodynamic Characteristics of Seven Symmetrical Airfoil Sections Through 180-Degree Angle of Attack for Use in Aerodynamic Analysis of Vertical Axis Wind Turbines,"* Sandia National Laboratories, Report No. SAND80-2114, 1981.

This report provides two types of data.

1.1 Tabulated Coefficient Data

The report includes a large table of C_L and C_D values across 0°–180° angle of attack, for multiple NACA profiles and Reynolds numbers. Not all of these values were measured directly. Sheldahl and Klimas combined the wind-tunnel results with numerical models to cover Reynolds numbers that were not tested in the wind tunnel. This is the data stored in the **Synthesized Data** folder.

1.2 Digitised Experimental Plot Data

The report also contains figures showing the actual wind-tunnel measurements. These measurements were only published as graphs, not as tables. The data points were recovered by digitising each figure using WebPlotDigitizer. This is the data stored in the **Experimental Data** folder.

2. Blackwell (1976) — Supplementary Source

Some additional NACA 0012 data was taken from:

> B. F. Blackwell, *"Wind Tunnel Performance Data for the Darrieus Wind Turbine*

with NACA 0012 Blades," Sandia National Laboratories, 1976.

This was used to cross-check and supplement the NACA 0012 results from the primary source.

3. Airfoiltools.com

Some reference data for comparison was taken from airfoiltools.com, an online database of airfoil performance data generated using XFOIL analysis.